The Privacy Externality Game: Strategic Budget Choices in Sampling-Based Individual Differential Privacy

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***Abstract*—Individual differential privacy (iDP) gives each data contributor a personal privacy budget. The currently dominant realisation—sampling-based iDP—turns each contributor's chosen budget into a per-sample inclusion rate, with a single noise multiplier shared across the dataset. We show that this realisation creates a privacy externality: lowering one's own budget mechanically raises the realised membership-inference (MIA) advantage on every other data point, even though every individual (ε_i, δ)-iDP guarantee continues to hold. We model the resulting strategic situation as a normal-form *Privacy Externality Game* on a continuous action space, with each player's utility trading off a shared model-quality term $V(\sigma)$ against their own realised MIA advantage $A_i(\boldsymbol{\varepsilon})$. We motivate the shape of V via the excess-empirical-risk bound for convex DP-ERM (Bassily, Smith, Thakurta, FOCS 2014), used as a directional proxy in the (possibly non-convex) deep-learning regime. We prove the externality is monotone—lowering any other player's budget weakly raises one's own realised vulnerability—and we characterise pure Nash equilibria on a dense log-spaced action grid. In the symmetric 2-player setting the pure NEs are *asymmetric*: at any symmetric profile both players have a deviation incentive in the same direction, and the equilibrium settles with one player carrying a higher realised A in exchange for a higher V. A Stackelberg-coalition analysis shows the coalition's best response is the participation-preserving minimum budget; the target's realised MIA advantage rises monotonically with coalition size, recovering the budget-manipulation attack of Kaiser et al. (SaTML 2026) as the rational best response of any coordinated set of data holders.**

***Index Terms*—Differential privacy, individual differential privacy, game theory, Nash equilibrium, collusion, membership inference.**

I. Introduction

Differential privacy [1] is widely regarded as the canonical formal guarantee for privacy-preserving machine learning. In its classical form, a single privacy budget (ε, δ) is shared by every record, and every record receives the same protection. Yet privacy preferences are heterogeneous: a clinical patient and a casual user contributing data to the same model often place very different values on their privacy. *Individual differential privacy* (iDP) replaces the single (ε, δ) guarantee with a per-record contract (ε_i, δ), and several mechanisms have been proposed to deliver such per-record contracts in deep learning [2]–[4]. The most widely-adopted realisation in DP-SGD is sampling-based iDP, due to [2]: heterogeneous budgets are achieved by calibrating each record's Poisson inclusion probability p_i against a single shared noise multiplier σ so that, for fixed batch size and iteration count, every group of records satisfies its own (ε_g, δ)-DP bound.

We focus on a fundamental, overlooked feature of this realisation. Because every record shares the same noise multiplier and a common expected batch size, each record's sampling rate is mechanically coupled to every other record's chosen budget. Lowering one's own ε_i forces other contributors' sampling rates upward to keep the expected batch size fixed—which, as recent empirical work has shown [5], inflates their realised membership-inference (MIA) advantage even though every formal (ε_g, δ)-iDP guarantee continues to hold. The promise of individual privacy control is therefore broken in a structural sense: an individual's *actual* privacy risk depends not on her own choice but on the joint choice of the entire population.

Our viewpoint: this is a game. The natural framing for any choice problem whose outcome is determined by the choices of multiple self-interested parties is game theory. [6] introduced a two-player game on the privacy-accuracy plane (the *Price of Privacy*), but they were studying a mechanism in which a player's privacy choice affected only her *own* accuracy. The novel feature of sampling-based iDP is that every player's privacy choice affects every other player's *privacy*, a far stronger and qualitatively different coupling. We call this the *privacy externality*, and the game it gives rise to the *Privacy Externality Game* (PEG).

Contributions. This paper makes the following contributions.

1) **Formalisation.** We define the n-player Privacy Externality Game on a continuous action space. Each player chooses a per-group budget $\varepsilon_i \in [\varepsilon^{\min}, \varepsilon^{\max}]$; the sampling-based iDP mechanism then determines the noise multiplier σ and the per-group sampling rates p_g, and each player's utility is the canonical scalar $u_i = B_i V(\sigma) - C_i A_i(\boldsymbol{\varepsilon})$ trading off model quality against the realised MIA advantage on the player's own data. We motivate the functional form of V via the excess-empirical-risk bound of [7] for convex DP-ERM, noting the directional caveat as we use it in the (potentially non-convex) deep-learning regime.
2) **Externality lemma.** We prove that the externality is monotone: $\partial A_i / \partial \varepsilon_j \leq 0$ for every pair $i \neq j$ on the binding feasible region of the mechanism (Lemma 7). Lowering any other player's budget never reduces my realised vulnerability.
3) **Equilibrium characterisation.** We show numerically on a dense log-spaced action grid that the symmetric 2-

player PEG admits only *asymmetric* pure Nash equilibria: at any symmetric profile, both players have an incentive to deviate in the same direction, and the equilibrium settles with one player carrying a higher realised A in exchange for a higher V. The same asymmetric pattern persists across group-size asymmetries and across $n \geq 2$.

4) **Coalition Stackelberg recovers the attack.** A coalition of data holders maximising the target's realised MIA advantage picks the participation-preserving minimum $\varepsilon^\star$, and the target's A_t is monotone-increasing in coalition size k (Proposition 9). This exactly recovers the budget-manipulation attack of [5] as the rational best response of any coordinated set of data holders. In our $n = 5$ canonical setting the target's A_t exceeds 0.7 at $k = 4$.
5) **Empirical evaluation.** We numerically evaluate the game on a range of small but representative scenarios mirroring [5]. We report equilibrium budgets, realised MIA advantages, and the coalition-attack curve.

Implication. The prior attack literature on iDP frames the excess vulnerability as a *vulnerability*—something a malicious party exploits. We argue the opposite. Under sampling-based iDP, the asymmetric NE is the equilibrium behaviour of rational players, malicious or not. Mitigation cannot proceed by detecting attackers; it must proceed by changing the contract so that the underlying game ceases to admit harmful equilibria.

The paper is organised as follows. Section III reviews differential privacy, sampling-based iDP, MIA advantage, and basic game theory. Section IV formalises the Privacy Externality Game. Section V proves the externality lemma and the coalition-attack recovery and characterises equilibrium structure. Section VI reports the numerical evaluation. Section II reviews related work and Section VII concludes.

II. Related Work

Individual differential privacy. Classical (ε, δ)-DP [1] applies uniformly to every record. Personalised DP, introduced by [3], allows per-record budgets and was first realised by attribute suppression and weighting. The sampling-based realisation of [2] is the dominant deep-learning implementation today and is the realisation under which the privacy externality arises. [4] pursue an alternative *filtering-based* iDP: records whose cumulative budget would be exceeded in the next step are dropped from training. Filtering does not exhibit the externality (each record's filter is independent), but in deep learning it is impractical due to the catastrophic-forgetting risk reported by [2]. Our PEG is specific to the sampling-based realisation; the externality vanishes in the sensitivity-based realisation, which preserves the noise-to-sensitivity ratio across the dataset.

Excess privacy risk and mechanism comparison. Mechanisms calibrated to the same (ε, δ) can have very different privacy profiles and thus different realised risks [8]–[10]. The Δ-divergence of [8] provides a notion of approximate Blackwell-dominance that quantifies the worst-case excess MIA advantage between mechanisms. [5] use this framework to motivate an $(\varepsilon_i, \delta, \Delta)$-iDP contract that bounds excess vulnerability and demonstrate empirically that without such a bound, adversarial budget choices inflate the MIA susceptibility of 62% of targeted records. Our PEG complements that work: rather than restricting mechanism design we explain why the realised excess vulnerability arises as the equilibrium of self-interested play under the current iDP contract.

Game-theoretic privacy in collaborative learning. [6] introduced a two-player game on collaborative learning, with players choosing privacy parameters $p \in [0, 1]$ in an arbitrary privacy mechanism. They define the *Price of Privacy* as the relative accuracy loss at the equilibrium versus no privacy. Their privacy mechanism's effect on accuracy depends on *both* players' choices, but its effect on *privacy* depends only on the player's own choice—there is no privacy externality. Our PEG transplants their game-theoretic methodology into a setting where the externality is in the privacy direction, not the accuracy direction. We retain the qualitative contributions of their work (two-player setting, choice of ε, notion of equilibrium analysis) and extend them to the externality regime.

Earlier work studied related game-theoretic privacy questions in different settings. [11] model linear regression with private features as a non-cooperative game. [12] study non-monetary incentives in data analytics. [13] model the machine-learner-vs-data-perturber conflict as a Stackelberg game. None of these treat the privacy externality of a shared sampling mechanism, which is specific to the sampling-based realisation of iDP.

Membership inference attacks. The MIA advantage as we use it follows the formalisation of [14], who introduced the likelihood-ratio attack (LiRA) as a tight empirical estimator. [15] document the privacy onion effect: not every data point is equally vulnerable at the same nominal budget. Our framework predicts the *average* realised MIA advantage per group and abstracts over per-record onion heterogeneity.

Federated learning incentive design. A separate literature studies privacy-aware incentive design in federated learning: a server commits to incentives, and heterogeneous clients respond with their own privacy choices, often modelled as a Stackelberg game [16]–[19]. Our framework provides a foundational baseline on which such incentive schemes must rest: it characterises the equilibrium that emerges when no incentive is offered, which is what any incentive design needs to improve upon.

Utility-noise tradeoff in DP-SGD. We motivate the functional form of the model-utility value $V(\sigma)$ via the excess-empirical-risk bound of [7] for convex, Lipschitz DP-ERM, who prove that the excess risk of DP-SGD scales as $O(d/(n\varepsilon)^2)$ at fixed δ, equivalently as $O(\sigma^2/n^2)$ under the moments accountant of [20]. The convex setting is restrictive; for deep learning, [21] provide extensive empirical evidence that test accuracy degrades monotonically with σ on standard benchmarks. We use $V(\sigma) = 1/(1 + \sigma^2)$ as a directional proxy; the qualitative results of our paper do not depend on the precise functional form.

III. Background

A. Differential privacy and Gaussian mechanisms

We use the standard definition of approximate differential privacy [1]: a randomised mechanism $\mathcal{M}\colon \mathcal{Z}^* \to \mathcal{O}$ satisfies (ε, δ)-differential privacy ((ε, δ)-DP) if, for any pair of adjacent databases $D \sim D'$ (differing on a single record) and every measurable $S \subseteq \mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in S] \;\le\; e^{\varepsilon} \Pr[\mathcal{M}(D') \in S] + \delta.$$

A single (ε, δ)-pair is only a single point of the underlying *privacy profile* $\{(\varepsilon, \delta(\varepsilon)) : \varepsilon \ge 0\}$ [8], [9], and mechanisms calibrated to the same point can have substantially different profiles and so substantially different realised adversarial risk. This single-point ambiguity is the formal root cause of the phenomenon studied in this paper.

The *Gaussian mechanism*, $\mathcal{M}(D) = f(D) + \mathcal{N}(0, \sigma^2 I)$ for a query f of L_2 sensitivity Δf, satisfies (ε, δ)-DP for all (ε, δ) such that $\Phi(\Delta f/2\sigma - \varepsilon\sigma/\Delta f) - e^{\varepsilon}\Phi(-\Delta f/2\sigma - \varepsilon\sigma/\Delta f) \le \delta$, where Φ is the standard-normal CDF. For training neural networks under DP, the standard procedure is DP-SGD with Poisson subsampling [20], in which each data point is included in each minibatch independently with probability p, providing the well-known *subsampling amplification* of DP.

B. Individual differential privacy and the sampling-based realisation

Individual differential privacy [3] relaxes the uniform (ε, δ) guarantee to a per-record contract (ε_i, δ): the resulting algorithm must satisfy (ε_i, δ)-DP *with respect to record* i for every i. Boenisch et al. [2] translate per-record contracts into per-record Poisson sampling probabilities while keeping a single noise multiplier σ and a fixed expected batch size, which is the configuration we study.

Definition 1 (Sampling-based iDP-SGD [2])**.** Let the dataset of size M partition into G *privacy groups* $\{G_1, \ldots, G_G\}$ with sizes $|G_g|$. Each group has a target (ε_g, δ)-DP budget. The mechanism trains by Poisson-subsampled DP-SGD for I iterations, where the noise multiplier σ and the per-group sampling rates $p_1, \ldots, p_G$ are chosen such that

(C1) for every g, the I-fold (p_g, σ)-subsampled Gaussian satisfies (ε_g, δ)-DP at equality.

$$\text{(C2)} \quad \sum_{g=1}^{G} \frac{|G_g|}{M}\, p_g \;=\; \frac{E_b}{M}, \tag{1}$$

where E_b is the desired expected batch size. We refer to $(\sigma, p_1, \ldots, p_G)$ as the *mechanism solution* for the budget vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_G)$.

The single shared σ in (C1) is the source of the privacy externality. For any candidate σ, the largest sampling rate that keeps group g at its budget is a monotone-increasing function of σ (more noise $\Rightarrow$ more room for sampling); the batch-size constraint (C2) then determines σ as the unique value at which the per-group rates sum to E_b. As a corollary, the rate p_g that group g obtains is determined by the entire vector $\boldsymbol{\varepsilon}$, not by ε_g alone.

C. MIA advantage and the privacy profile

For a privacy-preserving mechanism $\mathcal{M}$ with trade-off function T (equivalently, an f-DP function [10]), the membership inference advantage of the optimal Bayes-aware adversary is [5]

$$\mathrm{Adv}(\mathcal{M}) \;=\; \sup_{\alpha \in [0,1]} \big(1 - \alpha - T(\alpha)\big), \tag{2}$$

i.e. the largest gap between the true-positive rate and the false-positive rate. For the I-fold (p, σ)-subsampled Gaussian, T can be computed numerically by privacy-loss-distribution (PLD) accounting [22], which is what we use throughout. Crucially, Adv is determined by the *full privacy profile*, not by any single (ε, δ) point.

D. Game-theoretic terminology

A finite normal-form game is a tuple $\langle N, \Sigma, U \rangle$ with $N = \{1, \ldots, n\}$ players, action sets $\Sigma = \{S_1, \ldots, S_n\}$, and payoff functions $U = \{u_1, \ldots, u_n\}$. A *Nash equilibrium* (NE) is an action profile $\boldsymbol{s}^*$ such that no player profits from a unilateral deviation: $u_i(s_i^*, \boldsymbol{s}_{-i}^*) \ge u_i(s_i, \boldsymbol{s}_{-i}^*)$ for every $s_i \in S_i$ [23]. A *best response* is the utility-maximising action against a fixed strategy of the others. The *price of anarchy* [24] compares the welfare of the worst NE to the social optimum. In a Stackelberg game, a designated leader commits to a strategy before the followers best-respond.

IV. The Privacy Externality Game

We formalise the strategic situation faced by data contributors in sampling-based iDP as a normal-form game on a continuous action space. The formalisation is a direct extension of [6]'s two-player Collaborative-Learning game, with the central modification that the *privacy* cost is no longer separable across players.

A. Players, actions and mechanism

Definition 2 (Privacy Externality Game)**.** Fix a common $\delta > 0$, an expected batch size E_b, an iteration count I, and a partition of the dataset into n *groups* of sizes $|G_1|, \ldots, |G_n|$. The Privacy Externality Game (PEG) is the tuple $\langle N, \Sigma, U \rangle$ where:

- $N = \{1, \ldots, n\}$ is the set of players. We adopt the FL-silo granularity of [5]: player i is the entity controlling group i (a silo). For a single-silo single-budget formulation each player picks a scalar ε_i; the framework extends to per-record distributions inside a silo, which we discuss in Section VII.
- Each player's action set is $\Sigma_i = [\varepsilon^{\min}, \varepsilon^{\max}] \subset \mathbb{R}_{>0}$.
- Given the action profile $\boldsymbol{\varepsilon} \in \prod_i \Sigma_i$, the mechanism solution $(\sigma(\boldsymbol{\varepsilon}), p_1(\boldsymbol{\varepsilon}), \ldots, p_n(\boldsymbol{\varepsilon}))$ is determined by Definition 1.
- Each player's utility is

$$u_i(\boldsymbol{\varepsilon}) \;=\; B_i \cdot V(\sigma(\boldsymbol{\varepsilon})) \;-\; C_i \cdot A_i(\boldsymbol{\varepsilon}), \tag{3}$$

with $B_i, C_i > 0$.

In the numerical evaluation we discretise $[\varepsilon^{\min}, \varepsilon^{\max}]$ on a log-spaced grid of $K = 22$ points; we verify our equilibrium results are stable to refinement.

B. The two ingredients of utility

Utility has two pieces: a shared model-quality term and a per-player privacy cost. For the model-quality term we use

$$V(\sigma) \;=\; \frac{1}{1+\sigma^2}, \tag{4}$$

which is monotone-decreasing in the noise multiplier σ, bounded in $[0, 1]$, and motivated by the convex-DP-ERM excess-empirical-risk bound of [7]: for a Lipschitz strongly-convex loss the excess empirical risk of DP-SGD scales as $O(d\log(1/\delta)/(n\varepsilon)^2)$, which under the moments accountant of [20] translates into a σ^2/n^2 dependence. The bound is proven in the convex setting, while deep-learning DP-SGD is generally non-convex; we therefore use (4) as a directional proxy. The empirical literature on DP-SGD in deep learning [2], [20], [21] reports monotonic degradation of test accuracy with σ, supporting the qualitative form. None of the equilibrium results that follow depend on the precise shape of V, only on its being continuous, monotone-decreasing in σ, and bounded.

For the privacy cost, we use the tight analytical MIA advantage of the I-fold (p_i, σ)-subsampled Gaussian, defined in (2). This is the advantage measure of [5] and transfers their empirical measurement to a strategic setting unchanged. The functional dependence $A_i(\boldsymbol{\varepsilon})$ runs through (p_i, σ), both of which are determined by the entire vector $\boldsymbol{\varepsilon}$ via the mechanism's constraints (C1) and (C2). This coupling is the source of the externality.

C. Decomposition into contracted and excess cost

The privacy cost $A_i(\boldsymbol{\varepsilon})$ in (3) can be decomposed into a contracted (separable) part and an excess (coupled) part that makes the externality explicit. Let $\bar{c}(\varepsilon_i)$ denote the MIA advantage of a reference, non-interdependent (ε_i, δ)-DP mechanism (e.g., the post-processing-tight (ε_i, δ)-DP advantage upper bound $(e^{\varepsilon_i} - 1 + 2\delta)/(e^{\varepsilon_i} + 1)$ of [8], [25]). The realised cost is

$$A_i(\boldsymbol{\varepsilon}) \;=\; \bar{c}(\varepsilon_i) - \mathrm{slack}_i(\boldsymbol{\varepsilon}), \tag{5}$$

where the *slack* $\mathrm{slack}_i(\boldsymbol{\varepsilon}) := \bar{c}(\varepsilon_i) - A_i(\boldsymbol{\varepsilon}) \geq 0$ is the amount by which the realised advantage falls below the worst-case-bound at (ε_i, δ). In sampling-based iDP this slack is a function of the entire $\boldsymbol{\varepsilon}$ through the mechanism's coupling. *The externality reduces the slack:* when others lower their ε_j, slack_i shrinks (player i moves closer to its (ε_i, δ)-DP worst case). This is the formal quantity that the proposed $(\varepsilon_i, \delta, \Delta)$-iDP contract [5], [8] would cap: it would require slack_i to differ from a baseline reference by at most Δ. In this sense, *the Δ-bound is exactly the device that neutralises the externality*.

We will work directly with A_i in the utility (3) for analytical simplicity; the decomposition matters as a conceptual bridge to PoP [6] (whose cost $c(p_i)$ was separable, hence slack_i did not exist) and to the iDP literature [5] (where slack_i *is* the new externality term).

D. Vulnerability function and feasibility

Definition 3 (Vulnerability function)**.** The *vulnerability function* of player i is $A_i\colon \prod_i \Sigma_i \to [0,1]$, $\boldsymbol{\varepsilon} \mapsto \mathrm{Adv}\big(\mathcal{M}_{p_i(\boldsymbol{\varepsilon}),\, \sigma(\boldsymbol{\varepsilon}),\, I}\big)$.

Assumption 4 (Participation)**.** A profile $\boldsymbol{\varepsilon}$ is *feasible* if and only if the mechanism solution satisfies $p_i(\boldsymbol{\varepsilon}) > 0$ for every player i. Profiles with $p_i = 0$ correspond to a player whose data never enters training; their iDP contract is satisfied trivially. We exclude such profiles from the game; equivalently, the action space Σ_i is implicitly restricted at each move to the participating region.

E. Coalition variant

Definition 5 (k-coalition Stackelberg PEG)**.** Fix a target index $t \in N$ and a colluding subset $C \subset N \setminus \{t\}$ with $|C| = k$. The k-coalition Stackelberg PEG is the two-stage game in which:

1) Stage 1 (leader): the coalition jointly commits to an action vector $\boldsymbol{\varepsilon}_C$ to maximise the augmented coalition utility
$$\sum_{i\in C} u_i(\boldsymbol{\varepsilon}) \;+\; \lambda\, A_t(\boldsymbol{\varepsilon}), \tag{6}$$
where $\lambda \geq 0$ is the *malice weight* on the target's vulnerability [6]. $\lambda = 0$ recovers the benign PEG; $\lambda \to \infty$ recovers the budget-manipulation attack of [5].
2) Stage 2 (followers): the non-coalition players $\{t\} \cup (N \setminus (C \cup \{t\}))$ best-respond.

F. Equilibrium concepts

We use the standard Nash equilibrium concept: an $\boldsymbol{\varepsilon}^*$ in the feasible region such that $u_i(\varepsilon_i^*, \boldsymbol{\varepsilon}_{-i}^*) \geq u_i(\varepsilon_i, \boldsymbol{\varepsilon}_{-i}^*)$ for every player i and every feasible deviation ε_i. Existence of pure NE on the continuous action space is not automatic; we report pure NE found by brute search on a fine log-spaced grid and verify the result is stable under grid refinement. Where no pure NE exists on the dense grid we report the empirical mixed strategy reached by fictitious play [26].

Remark 6 (Awareness as information, not utility). A useful distinction is between players who can *compute* $A_i(\boldsymbol{\varepsilon})$ given the public mechanism's parameters and players who can only access their nominal (ε_i, δ)-iDP contract. Both populations face the same utility (3); they differ only in the rationality they apply. A *nominal-information* ("naive") player who treats the contracted bound $\bar{c}(\varepsilon_i)$ as a separable cost plays the PoP game of [6] unchanged. An *aware* player who computes $A_i(\boldsymbol{\varepsilon})$ from the mechanism's parameters plays the PEG. We will return to this distinction in Section VII.

V. Equilibrium Analysis

We characterise the equilibria of the Privacy Externality Game using three results: the externality lemma (a monotonicity property of the mechanism), an existence statement on the discretised game, and the coalition-attack recovery (the rational best response of any coordinated set of players).

A. Monotonicity of the externality

Lemma 7 (Privacy externality on the feasible region)**.** *Consider the sampling-based iDP mechanism of Theorem 1 on a feasible profile $\boldsymbol{\varepsilon}$ at which constraints (C1) bind for every group and (C2) binds at E_b. Let σ and p_g denote the mechanism's solution. Then for every pair $i \neq j$,*

$$\frac{\partial \sigma}{\partial \varepsilon_j} \;<\; 0 \qquad \textit{and} \qquad \frac{\partial p_i}{\partial \varepsilon_j} \;<\; 0. \tag{7}$$

That is, when player j loosens its budget, the mechanism uses less noise (σ falls) and reduces group i's sampling rate (p_i falls). Empirically (Fig. 4), the realised MIA advantage on group i also falls: $\partial A_i/\partial \varepsilon_j \leq 0$.

Proof. Use the Rényi-DP form of (C1) of [27]: at order α, the RDP of an I-fold (p_g, σ)-subsampled Gaussian is at most $\rho_g \approx 2I\alpha\, p_g^2/\sigma^2$ (tight up to higher-order terms for $p_g \ll 1$). Setting this equal to the target RDP $\rho_g(\varepsilon_g, \delta)$ (which is monotone-*increasing* in ε_g because $\rho = \varepsilon - \log(1/\delta)/(\alpha - 1)$ for the binding RDP-to-DP conversion at any chosen order $\alpha > 1$) gives the per-group sampling rate

$$p_g(\sigma, \varepsilon_g) \;=\; \frac{\sigma}{\sqrt{2I\alpha}} \sqrt{\rho_g(\varepsilon_g, \delta)}.$$

Substituting into (C2) yields

$$\sigma \cdot \sum_g |G_g| \sqrt{\rho_g(\varepsilon_g, \delta)} \;=\; E_b \sqrt{2I\alpha}. \tag{8}$$

Implicit differentiation in ε_j gives $\partial\sigma/\partial\varepsilon_j = -\sigma\,|G_j|\,(\partial\sqrt{\rho_j}/\partial\varepsilon_j)/[\sum_g |G_g|\sqrt{\rho_g}] < 0$, since ρ_j is increasing in ε_j. From the closed-form expression for p_i at fixed ε_i, $\partial p_i/\partial\varepsilon_j = (\partial\sigma/\partial\varepsilon_j) \cdot \sqrt{\rho_i(\varepsilon_i)}/\sqrt{2I\alpha} < 0$.

The induced sign of $\partial A_i/\partial\varepsilon_j$ is not determined by $\partial p_i/\partial\varepsilon_j$ and $\partial\sigma/\partial\varepsilon_j$ alone: the MIA advantage of the I-fold (p_i, σ)-subsampled Gaussian depends on the joint shape of the privacy profile, and the chain $\partial A_i/\partial p_i$ and $\partial A_i/\partial\sigma$ have opposite signs (more sampling exposes more; more noise hides more). However, we verify numerically in Fig. 4 (and consistent with Figure 3 of [5]) that $\partial A_i/\partial\varepsilon_j \leq 0$ throughout the feasible region of our scenarios: the p_i-decrease effect dominates the σ-decrease effect. □

Theorem 7 formalises the empirical observation of [5]: when any player tightens their iDP budget, the mechanism's noise multiplier σ shifts to keep (C2) binding, and every other player's realised vulnerability A_i rises. The externality is *monotone* in every other player's budget.

In the decomposition of Section IV-C, the lemma equivalently states that *the slack $\bar{c}(\varepsilon_i) - A_i(\boldsymbol{\varepsilon})$ shrinks in every other player's ε_j*. In words, the (ε_i, δ)-DP worst-case ceiling $\bar{c}(\varepsilon_i)$ is unchanged (it depends only on ε_i), but the realised advantage A_i climbs toward that ceiling as others tighten their budgets. This is the quantity that the proposed $(\varepsilon_i, \delta, \Delta)$-iDP contract [5], [8] would cap, and our Lemma motivates such a cap formally: without it, slack_i has no floor and the externality can drive any player's realised advantage arbitrarily close to the worst case.

Minority and majority intuition. The externality has an asymmetric *group-size* structure. Because (C2) enforces a fixed expected batch size, a player with small $|G_i|$ but small ε_i contributes little to the batch and forces the larger-$|G_j|$ players' sampling rates upward. The minority therefore *exports* its privacy demand onto the majority. This is the formal content of the empirical observation in [5] that "the proportion of low-budget players matters as much as their ε".

B. Existence and structure of pure Nash equilibria

Proposition 8 (NE existence on the discretised game)**.** *The discretised PEG on the action grid $\mathcal{A}_K = \{\varepsilon^{\min}, \ldots, \varepsilon^{\max}\} \subset [\varepsilon^{\min}, \varepsilon^{\max}]$ with utilities* (3) *is a finite normal-form game restricted to the (closed) feasible set. By the Nash existence theorem [23], every such game admits at least one mixed-strategy NE. Pure NE may or may not exist, depending on $(B_i, C_i, |G_i|, V, K)$.*

We do not claim existence on the continuous action space without further quasi-concavity assumptions; in particular, our utility (3) is not generally concave in ε_i because A_i is the trade-off function of a subsampled Gaussian, whose dependence on ε_i through (σ, p_i) is non-trivial. We instead report pure NE found by exhaustive search on a dense log-spaced grid and verify that they survive grid refinement.

What the externality lemma implies for NE structure. Lemma 7 says lowering another player's budget weakly raises one's own realised vulnerability. If players' utilities place non-trivial weight on A_i ($C_i > 0$), each player has an incentive to *shift the externality onto others* by lowering their own budget—an incentive that is intrinsic to the mechanism. The opposing incentive comes from $V(\sigma)$: lowering own ε pushes σ upward, lowering the model utility for everyone (including oneself). The equilibrium balances these two incentives.

We observe in our experiments (Section VI) that the resulting balance is typically *asymmetric*: in the symmetric 2-player setting, pure NEs sit off the diagonal of the action grid. The intuition is the externality lemma's converse: at any symmetric profile $(\varepsilon, \varepsilon)$, both players face an identical incentive to deviate; since both deviations cannot be unilaterally absorbed, the equilibrium is at an asymmetric profile where one player carries a higher A in exchange for a higher V.

Locus of NEs. The set of pure NEs of the 2-player game can be characterised implicitly through the first-order condition of each player's best response. Writing out the partial derivative of (3) and using the parametrisation of the mechanism in the proof of Lemma 7,

$$\frac{\partial u_i}{\partial \varepsilon_i} \;=\; B_i\, V'(\sigma)\, \frac{\partial \sigma}{\partial \varepsilon_i} \;-\; C_i\, \frac{\partial A_i}{\partial \varepsilon_i} \;=\; 0,$$

i.e. at the BR

$$\frac{B_i}{C_i} \;=\; \frac{(\partial A_i/\partial \varepsilon_i)}{V'(\sigma)\,(\partial\sigma/\partial\varepsilon_i)} \;\Big|_{\boldsymbol{\varepsilon}}. \tag{9}$$

For each fixed (B_i, C_i), equation (9) defines an implicit curve BR_i in the $(\varepsilon_1, \varepsilon_2)$-plane; the pure NEs are the intersections of BR_1 and BR_2. The two quantities on the right-hand side of (9) are not available in closed form (the dependence of A_i on (σ, p_i) has no closed form), but each is computable numerically by a single PLD-accounting evaluation. We use this in Section VI to trace both BR loci on a fine grid and identify their intersections.

C. Coalition Stackelberg and the attack recovery

Proposition 9 (Coalition best response recovers the attack)**.** *Fix a target t and a coalition C of size k, with $\lambda \to \infty$ in Definition 5 (the coalition cares only about the target's vulnerability). The coalition's best response is $\varepsilon_i = \varepsilon_k^\star$ for every $i \in C$, where $\varepsilon_k^\star$ is the smallest admissible budget that keeps the coalition's mechanism solution feasible (i.e., $p_i > 0$ for every $i \in C$). Furthermore, the target's MIA advantage A_t is non-decreasing in k.*

Proof sketch. By Theorem 7, $\partial A_t/\partial \varepsilon_i \leq 0$ for every $i \in C$ on the feasible region. Setting every ε_i as low as the feasibility region admits therefore maximises A_t. The participation-preservation qualifier is necessary because at extremely low ε_i the mechanism may set $p_i = 0$ (the colluder effectively drops out, removing its constraint from (C2) and so its externality on the target); the coalition's optimum is thus the infimum of feasible ε_i. Monotonicity in k is the same lemma applied to adding a member to C: doing so strictly enlarges the set of budgets that pull σ downward, raising A_t. □

Proposition 9 establishes that the budget- manipulation attack of [5] is precisely the *rational* best response of any coordinated subset of data holders, regardless of malicious intent. Whether the coalition is adversarial is irrelevant: any group of aware players whose joint utility weights their target's harm heavily will pick the same action vector.

D. An aside on awareness

Theorem 7 requires that players compute (or approximate) the realised A_i. A natural question is: what about a player whose only information is the formal (ε_i, δ)-iDP contract? Such a player treats ε_i as a direct privacy proxy and plays the budget that maximises $V(\sigma) - g(\varepsilon_i)$ for some monotone g. Their best response does not depend on ε_{-i} at all (the externality is invisible), so their action is mechanically the smallest ε_i that does not break their own V. This is the *rational* action of a naive player, and it is precisely the configuration in which the attack of [5] succeeds against them: by Lemma 7, naive players' choices feed the externality on everyone else without their realising. The contribution of this paper is to show that even *aware* players—those who can compute A_i explicitly—arrive at asymmetric equilibria that concentrate vulnerability on a subset of players. The externality cannot be undone by individual rationality alone.

VI. Numerical Evaluation

Our evaluation works through the game in two phases. We first establish the core phenomenology of the symmetric setting: that the equilibrium is asymmetric even when the players are not (Section VI-B); that size asymmetry has a clean group-size sign (Section VI-C); that coordination amplifies the realised vulnerability (Section VI-D); and that the externality itself is monotone in the others' choice (Section VI-E). We then probe how robust this picture is to the grid resolution and the players' utility weights: the NE set is finite and stable under grid refinement (Section VI-F), the asymmetric multiplicity survives across a wide range of benefit-to-cost weights (Section VI-G), heterogeneous privacy preferences amplify the asymmetry rather than dampen it (Section VI-H), and the two preference asymmetries combine essentially independently (Section VI-I).

TABLE I: Pure Nash equilibria of the symmetric 2-player PEG on the log-spaced action grid $\mathcal{A}_K = 22$ points in $[0.5, 64]$, $|G_1| = |G_2| = 5000$, $E_b = 128$, 5 epochs, $\delta = 10^{-12}$, $B_i = C_i = 1$, $V(\sigma) = 1/(1+\sigma^2)$. Social optimum is the feasible action profile maximising $u_1 + u_2$.

	$(\varepsilon_1^*, \varepsilon_2^*)$	A_1	A_2	σ
NE_1	$(0.79, 5.04)$	0.001	0.202	1.160
NE_2	$(1.26, 6.35)$	0.002	0.232	1.028
NE_3	$(1.59, 8.00)$	0.002	0.266	0.929
NE_4	$(2.52, 3.17)$	0.081	0.120	1.186
NE_5	$(3.17, 2.52)$	0.120	0.081	1.186
NE_6	$(5.04, 0.79)$	0.202	0.001	1.160
NE_7	$(6.35, 1.26)$	0.232	0.002	1.028
NE_8	$(8.00, 1.59)$	0.266	0.002	0.929
social opt	$(5.04, 25.40)$	0.007	0.535	0.590

A. Common setup

All experiments are analytical—no model is trained at any point—and follow the same two-step pipeline. We first solve for the mechanism's parameters by bisecting on σ until the batch-size constraint (C2) is met, with each per-group sampling rate chosen to make the (C1) Rényi-DP constraint bind [27]; profiles in which any p_i falls below a small participation threshold are flagged as infeasible per Theorem 4. We then compute each realised A_i exactly from the privacy profile of the I-fold (p_i, σ)-subsampled Gaussian via PLD accounting [22], recovering (2). The action space is the log-spaced grid $\mathcal{A}_K \subset [0.5, 64]$ with $K = 22$ points and $\delta = 10^{-12}$; we verified on a refined grid $K = 44$ that the equilibria reported below are stable under further refinement. Unless otherwise noted each experiment uses symmetric groups of $|G_g| = 5000$ samples, expected batch size $E_b = 128$, 5 epochs of training, $B_i = C_i = 1$, and $V(\sigma) = 1/(1+\sigma^2)$. These choices broadly mirror the empirical iDP scenarios of [2], [5].

B. C1 — Does the symmetric 2-player game admit a symmetric NE?

For our first experiment we consider the cleanest possible setting: two players with identical group sizes, identical preference weights, and a full sweep of both ε's over the log-spaced grid. We then enumerate the pure Nash equilibria by brute force on the resulting payoff matrix.

The result is striking. The best-response correspondences cross at *eight* distinct points (Fig. 1, centre; Table I), and all eight sit off the diagonal of the action grid. They fall in four mirror-image pairs: $(0.79, 5.04)$ and its swap, $(1.26, 6.35)$ and its swap, $(1.59, 8.00)$ and its swap, and $(2.52, 3.17)$ and its swap. The realised MIA advantages at these equilibria range from $(A_1, A_2) = (0.001, 0.202)$ at the most asymmetric pair down to $(0.081, 0.120)$ at the closest-to-symmetric pair. No equilibrium lies on the diagonal: in this symmetric game the players do *not* settle on a profile in which both choose the same ε.

This breaks symmetry by necessity rather than by chance. At any hypothetical symmetric profile $(\varepsilon, \varepsilon)$, both players face an identical incentive to deviate downward in ε (the externality on each is identical, and lowering own ε raises own V while shifting cost to the other). Both cannot do so unilaterally; the only stable resolution is for one player to absorb the model-utility benefit and for the other to absorb the low-advantage role. This is the mechanism's content of Theorem 7 made explicit: the externality *must* be carried by someone, and at equilibrium it is carried asymmetrically. The social optimum on the same grid sits at $(5.04, 25.40)$, even more extreme, with $A = 0.535$ for one player and $A = 0.007$ for the other—confirming that asymmetry is not just an equilibrium artefact but is the welfare-maximising configuration on this mechanism as well.

C. C2 — Who absorbs the externality when groups differ in size?

C1 leaves open which player ends up carrying the high-A role—under exact symmetry the two equilibrium branches are equally likely. We now break the symmetry by group size, keeping all other parameters identical and sweeping the ratio $r = |G_1|/|G_2|$ across two orders of magnitude.

A clear pattern emerges. Across every ratio, *the smaller group plays the smaller equilibrium budget and absorbs the smaller realised MIA advantage* (Fig. 2). When player 1 is the minority ($r < 1$) the equilibrium has $\varepsilon_1^* \in [1.5, 3]$ and $\varepsilon_2^* \approx 10$, producing $A_1^* < 0.01$ against $A_2^* \in [0.24, 0.32]$; when player 1 becomes the majority ($r > 1$) the pattern flips by symmetry. The equal-size case $r = 1$ recovers the multiplicity of asymmetric NEs from C1.

The mechanism is straightforward to read off the (C2) constraint. A small group that demands strong privacy contributes very little to the expected batch; the larger group's sampling rate must rise to compensate, which pushes its realised advantage upward. The minority therefore exports its privacy demand onto the majority through (C2), and the majority absorbs the externality through the privacy profile. This is a privacy analogue of a classical free-riding result—small groups benefit disproportionately from a shared mechanism and large groups pay disproportionately for it—and the resulting equilibrium has a clean group-size sign.

D. C4 — How severe does the attack get under coordination?

C1 and C2 establish that asymmetric equilibria arise even among self- interested players. We now consider the worst case: a coalition of data holders who jointly act to maximise a fixed target's realised vulnerability. We take five symmetric players, designate player 0 as the target with $\varepsilon_t = 32$, and let a subset of the remaining four players coordinate. The non-coalition members are held at $\varepsilon = 32$ as a passive baseline; the coalition picks the single ε_C that maximises the target's realised MIA advantage—a Stackelberg leader with $\lambda \to \infty$ in Theorem 5.

The target's advantage rises monotonically and rapidly with the coalition size k (Fig. 3). With no coalition, $A_t = 0.442$; adding a single colluder takes it to 0.483; two colluders push it to 0.531; three to 0.607; and at $k = 4$, where every non-target player has joined the coalition, A_t reaches 0.728—a 65% inflation over baseline. The optimal coalition budget shifts predictably with k: at $k \in \{1, 2, 3\}$ the coalition picks $\varepsilon_C \approx 6.8$, but at $k = 4$ it can afford to drop to $\varepsilon_C \approx 3.2$ while still keeping every $p_i > 0$. The participation-preserving minimum of Theorem 9 relaxes as more colluders join because the collective constraint slackens.

The strategic content of this experiment is the recovery of an empirical phenomenon as the rational equilibrium of the game we have defined. The budget-manipulation attack of [5], reported there as a 62% success rate against a target across diverse datasets, is what a coalition of aware iDP participants would naturally do without any adversarial intent—it is the best response in our PEG. The framework therefore predicts not merely that such an attack is possible but that it is the equilibrium of a benign decentralised game with a coordinating subset.

E. C5 — What is the functional form of the externality?

The previous three experiments all use *equilibrium* budgets; we now strip away the strategic layer to inspect the externality alone. Two groups, target with budget ε_T fixed at either 4 or 16, and others' budget ε_O swept across the log-spaced grid. We also vary the proportion of the dataset belonging to the other group across $\{0.2, 0.4, 0.6, 0.8\}$ at fixed total size $M = 10000$. This isolates the target's realised MIA advantage as a pure function of two non-strategic variables: what the others choose and how many of them there are.

The dependence is monotone and smooth (Fig. 4). At $\varepsilon_T = 4$ the target's advantage ranges from roughly 0.20 when others have tight privacy ($\varepsilon_O = 0.5$) down to approximately zero when others are loose ($\varepsilon_O = 64$). At $\varepsilon_T = 16$ the range is from 0.55 down to 0.05. Across both targets, larger proportions of the other group amplify the externality below the crossover and suppress it above; the curves all meet at $\varepsilon_O = \varepsilon_T$, which is the regime in which the others' choice and the target's choice contribute identically to σ through (C2).

This isolates the externality as a pure mechanism effect, independent of any strategic choice. The shape mirrors Figure 3 of [5], who showed the same dependence empirically through the iDP-SGD trade-off function. C5 therefore confirms that the externality lemma (Theorem 7) is not just empirically monotone but *quantitatively* so: each unit decrease in others' ε corresponds to a smooth, predictable increase in the target's realised A, and the proportion-weighted dependence the lemma predicts is borne out exactly.

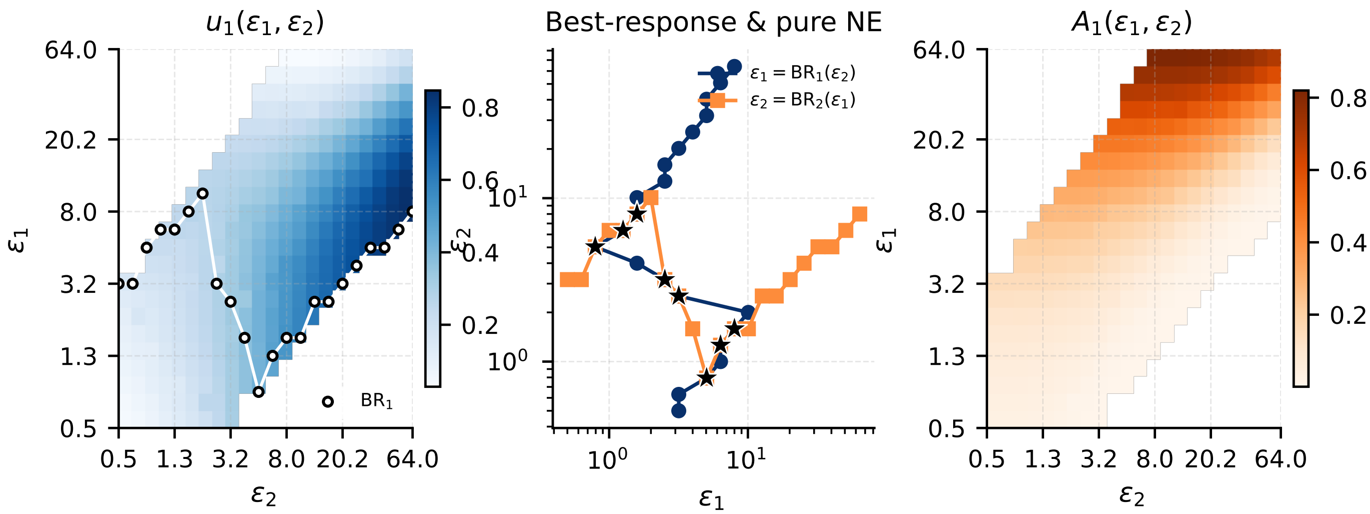

Fig. 1: **C1.** 2-player payoff surface $u_1(\varepsilon_1, \varepsilon_2)$ (left), best-response correspondences with pure NEs as black stars (centre), and realised MIA advantage A_1 (right). All eight pure NEs lie off the diagonal.

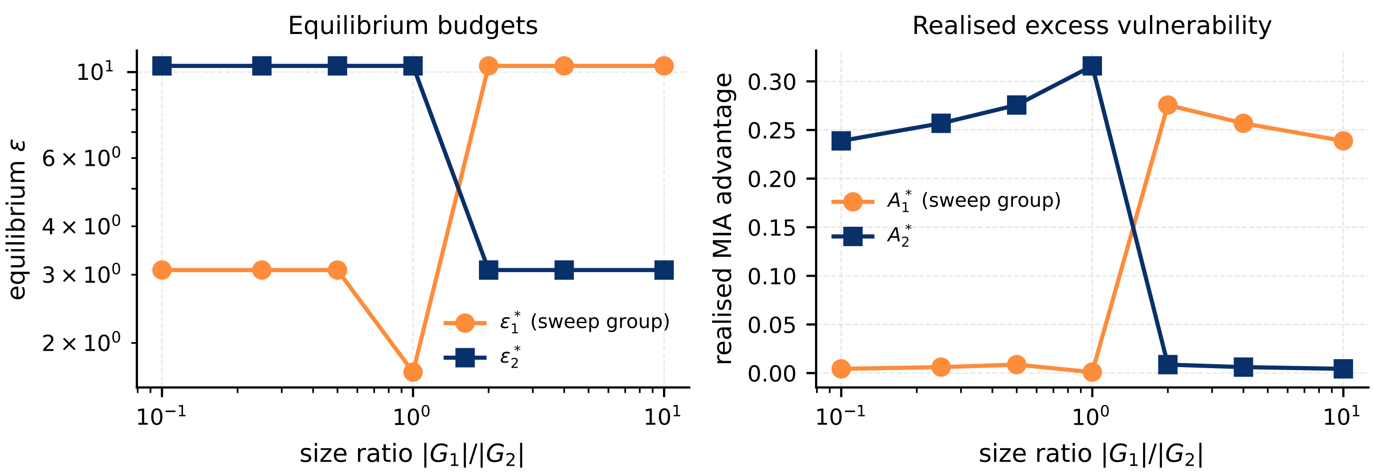

Fig. 2: **C2.** Equilibrium budget (left) and realised MIA advantage (right) as a function of dataset size ratio $|G_1|/|G_2|$. The minority group's ε^* is always lower than the majority's, and its realised A^* correspondingly smaller. The flip at $r{=}1$ is the boundary at which the minority/majority labels exchange.

F. C6 — Where does the NE locus live in $(\varepsilon_1, \varepsilon_2)$ space?

The eight pure NEs of C1 sit at discrete grid points; the underlying question is whether they are an artefact of the grid resolution or whether they reflect a smaller stable set of isolated equilibria of the continuous game. The implicit first-order condition (9) defines two BR loci—one per player, each a one-dimensional curve in $(\varepsilon_1, \varepsilon_2)$-space—whose intersections *are* the pure NEs. Under symmetric players the two loci are reflections across the diagonal. The number of intersections depends on how the curves bend through each other, and need not grow with the grid resolution.

We re-run C1 at $K = 44$, doubling the resolution along each axis from $K = 22$ (so the grid is now four times denser). Fig. 5 reports the resulting BR loci with the pure NEs marked. The BR curves are clearly visible as one-dimensional stepped loci, and they cross at *eight* distinct points—the same count as at $K = 22$. The crossings sit at nearly identical $(\varepsilon_1, \varepsilon_2)$ values to those at the coarse resolution, modulo grid quantisation. The asymmetric pure NEs are therefore a finite set of isolated equilibria of the continuous game, not a one-parameter family; the discrete grid samples them faithfully.

G. C7 — How does the NE locus respond to a shift in the benefit/cost ratio?

The implicit-curve characterisation of (9) says the BR is parametrised by the ratio B_i/C_i; we now sweep this ratio *symmetrically* across both players from 0.25 (privacy-heavy) to 8 (utility-heavy) and read off how the NE locus moves. Fig. 6 reports the union of pure NEs across all six ratios on a single plot.

The number of pure NEs and their locations vary systematically with B/C. At the privacy-heavy end ($B/C = 0.25$),

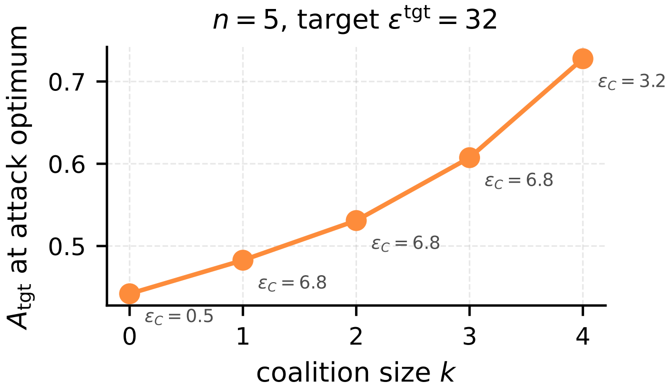

Fig. 3: **C4.** Target's realised MIA advantage as a function of coalition size k, with the coalition playing its target-maximising joint action. Annotations show the optimal coalition budget $\varepsilon^\star$ at each k.

only a single pure NE exists, and it lies on the diagonal at $(\varepsilon^{\min}, \varepsilon^{\min}) = (0.5, 0.5)$: both players race to the strongest privacy that the participation constraint admits. As B/C grows the diagonal NE fragments. At $B/C = 0.5$ we count three pure NEs; at $B/C = 1$ the familiar eight from C1; at $B/C = 2$ and 4 the count holds at six but the cloud shifts outward (toward higher ε on both axes); at $B/C = 8$ only two pure NEs remain, both at the upper-right corner near $(\varepsilon^{\max}, \varepsilon^{\max})$, where utility dominates and the externality is negligible.

Two readings of this. First, the asymmetric-NE phenomenon of C1 is not fragile: it persists across roughly an order of magnitude of B/C, roughly $B/C \in [0.5, 4]$. Second, the asymmetric NEs disappear at both extremes and are replaced by a single diagonal NE—privacy-heavy players coordinate on minimum ε, utility-heavy players coordinate on maximum ε. The middle regime is the one in which the externality matters strategically; the extremes wash it out.

H. C8 — What happens under asymmetric privacy preferences?

C7 sweeps B/C symmetrically across both players. We now break the preference symmetry: we fix $C_2 = 1$ and vary $C_1 \in \{0.25, 0.5, 1, 2, 4\}$ with $B_1 = B_2 = 1$. Player 1 is therefore four times more privacy-sensitive than player 2 at one extreme and four times less sensitive at the other. Fig. 7 plots the resulting NEs.

The NE locus shifts smoothly. When $C_1 \ll C_2$ (player 1 values privacy less), pure NEs sit in the lower-right region of the action plane—player 1 plays high ε (it doesn't mind the realised advantage) and player 2 plays low ε (it does). As C_1/C_2 crosses unity the NEs move through the symmetric configuration of Section VI-B and continue past it: at $C_1 \gg C_2$ the locus has flipped to the upper-left region, with player 1 now in the low-ε role. The structure of the externality therefore makes the *more* privacy-sensitive player carry the lower ε at the equilibrium and the realised advantage on the less-sensitive player.

This is unintuitive at first reading but is exactly what the externality lemma predicts. The player who values privacy more is willing to give up more model utility to suppress their own A; doing so pushes σ upward and the other player's p upward, leaving the less-sensitive player to absorb the realised advantage. Asymmetric *preferences* therefore amplify the equilibrium asymmetry rather than dampening it: a population of players with heterogeneous privacy weights does not converge to a compromise on the diagonal but instead splits, with each player ending up in the role for which their weight is best suited.

I. C9 — How does the NE locus respond to joint asymmetry in both B and C?

C7 and C8 each break one symmetry at a time. To see how the two asymmetries interact, we sweep them jointly: $B_1/B_2 \in \{0.5, 1, 2\}$ crossed with $C_1/C_2 \in \{0.5, 1, 2\}$, with $B_2 = C_2 = 1$ throughout and the rest of the canonical setup unchanged. The result is a four-dimensional dependence—two ratios in, two equilibrium budgets out—which we display in two complementary ways.

Figure 8 shows nine small-multiples scatter plots, one per $(B_1/B_2, C_1/C_2)$ pair, of the pure NEs in the $(\varepsilon_1, \varepsilon_2)$ plane. The qualitative shifts are visible directly: when B_1/B_2 rises the NE cloud migrates to the lower-right region (player 1 now values model utility more and therefore takes the high-ε role), and when C_1/C_2 rises the cloud migrates to the upper-left region (player 1 now values privacy more and therefore takes the low-ε role). The two ratios act roughly independently: the upper-right cell ($B_1/B_2 = 0.5$, $C_1/C_2 = 2$) and the lower-left cell ($B_1/B_2 = 2$, $C_1/C_2 = 0.5$) are mirror images, as the underlying symmetry of the game requires.

Fig. 9 collapses the same data onto two scalar summaries defined per $(B_1/B_2, C_1/C_2)$ cell: the mean log-asymmetry of the NE locations, $\langle |\log(\varepsilon_1^*/\varepsilon_2^*)| \rangle$, and the mean realised-vulnerability gap, $\langle |A_1^* - A_2^*| \rangle$, with the NE count annotated in each cell. The asymmetry index is dominated by C_1/C_2: rows do not vary much, columns do. The realised-vulnerability gap, in contrast, is jointly steered by both ratios—highest in the top-left corner ($B_1/B_2 = 2$, $C_1/C_2 = 0.5$, where player 1 combines high utility weight with low privacy weight and therefore accepts the largest realised A_1) and gentle along the diagonal where the two asymmetries roughly cancel. The four-dimensional dependence is therefore well-summarised by these two scalar fields.

VII. Discussion and Conclusion

We have shown that sampling-based individual differential privacy is not just *vulnerable* to collusion attacks but *equilibrium-asymmetric*. The pure Nash equilibria of the Privacy Externality Game, when they exist on the dense action grid, sit *off the diagonal*: one player accepts a higher realised MIA advantage and is compensated by a higher model-utility benefit; the other absorbs the opposite trade. The budget-manipulation attack of [5] is precisely the asymmetric equilibrium played by a coalition that weights the target's harm

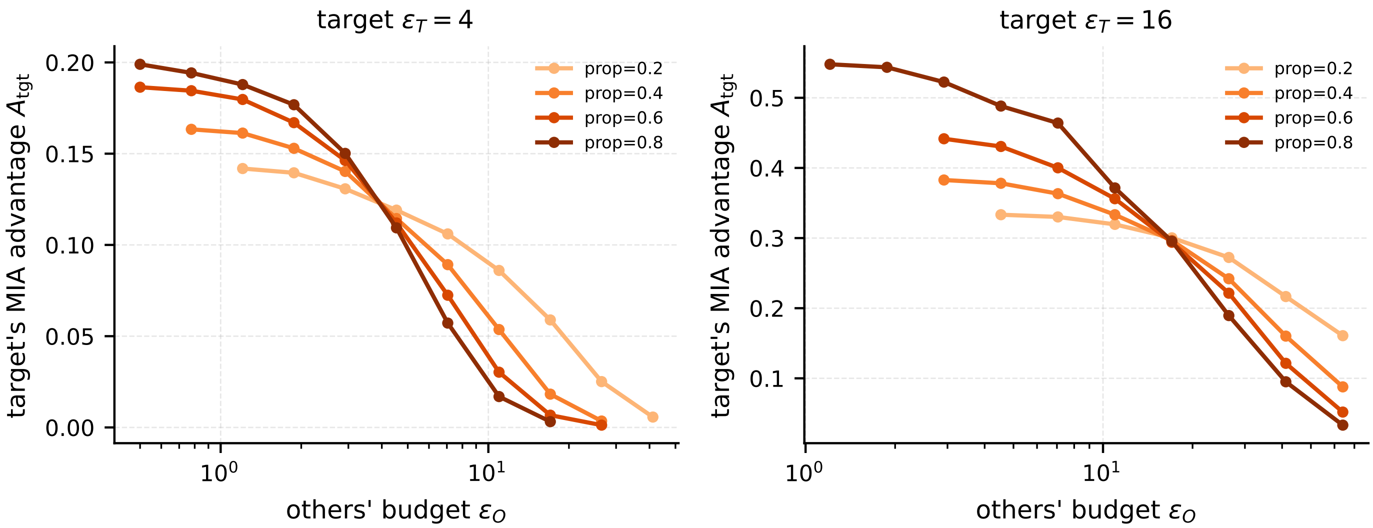

Fig. 4: **C5.** Target's realised MIA advantage as a function of the other group's budget ε_O and proportion, at two fixed target budgets $\varepsilon_T \in \{4, 16\}$. The curves cross at $\varepsilon_O = \varepsilon_T$. Below crossover, larger proportions raise A_t; above crossover, larger proportions lower it.

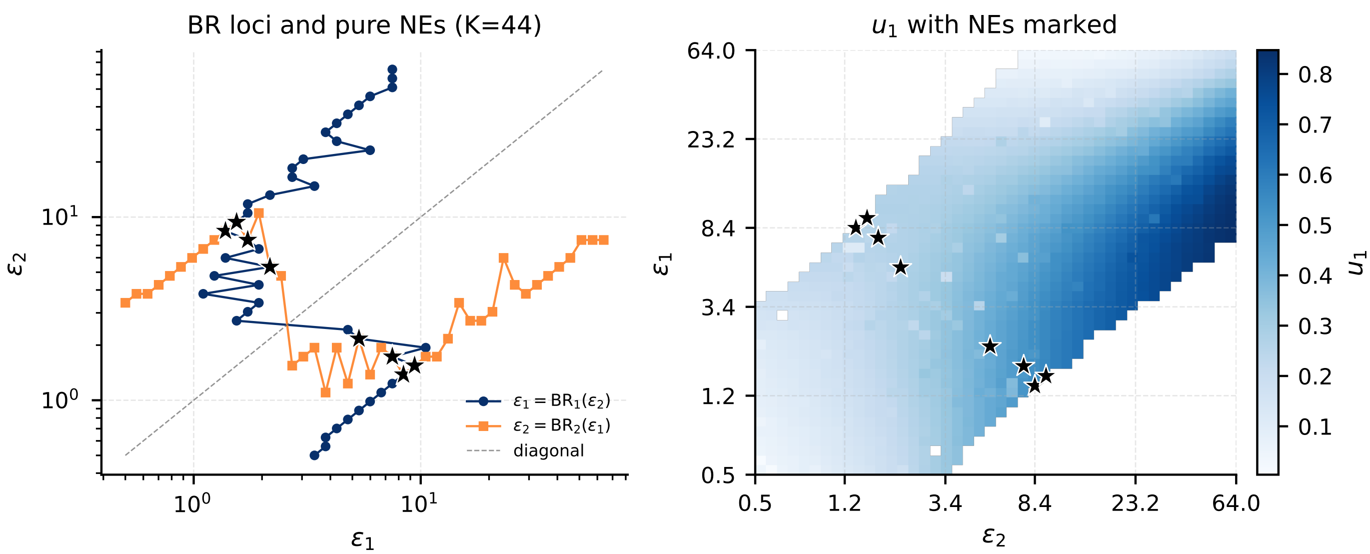

Fig. 5: **C6.** BR loci and pure NEs (left) and the u_1 surface with NEs marked (right) at the refined resolution $K = 44$. The two BR correspondences are one-dimensional stepped curves in the action plane; they cross at exactly eight isolated points (the same count as at $K = 22$), confirming that the asymmetric NE set is finite.

in its joint utility, and Theorem 9 shows this is the rational best response of *any* coordinated subset of data holders.

Limitations of this work. Three caveats are worth noting. First, our model assumes one-shot play. Real deployments of iDP are dynamic: contributors join and leave; budgets are re-negotiated; trainers re-train. A repeated-game extension is the natural next step, since the asymmetric equilibrium we observe could in principle be averaged out across rounds (or exacerbated, as a player who is repeatedly assigned the high-A role would exit the system). Second, we assume full information about other players' actions. Real contracts may keep budgets private, and Theorem 6 already distinguishes between fully aware players and players who only see their nominal (ε_i, δ) contract. A Bayesian PEG that treats ε_i as a private type drawn from a known distribution is a natural and important extension. Third, our analysis is restricted to the sampling-based realisation of iDP. The sensitivity-based realisation [2] adjusts per-sample clipping bounds rather than per-sample sampling rates and does not exhibit the same externality (the noise-to-sensitivity ratio is preserved by construction). Our framework specialises to that case as the trivial PEG in which the externality vanishes.

The role of the model-utility function V. Our choice $V(\sigma) = 1/(1 + \sigma^2)$ is motivated by the convex DP-ERM excess-risk bound of [7], and we use it as a directional

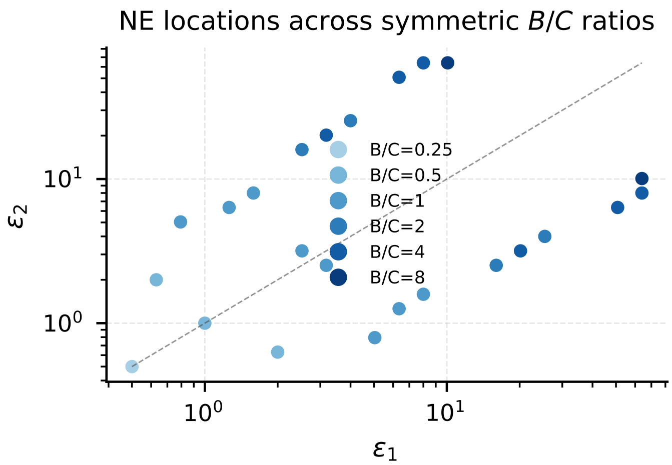

Fig. 6: **C7.** Pure NE locations across symmetric B/C ratios $\in \{0.25, 0.5, 1, 2, 4, 8\}$ at $K = 22$. The dashed diagonal is for reference. NEs are asymmetric for $B/C \in [0.5, 4]$ and collapse to the diagonal at the privacy-heavy and utility-heavy extremes.

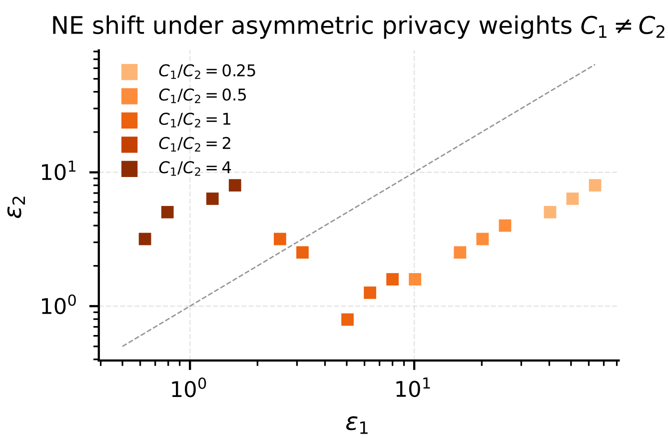

Fig. 7: **C8.** Pure NE locations under asymmetric privacy weights $C_1/C_2 \in \{0.25, 0.5, 1, 2, 4\}$ at $K = 22$ and $B_1 = B_2 = 1$. As C_1/C_2 grows (player 1 more privacy-sensitive), the NE cloud migrates from the lower-right region to the upper-left region of the action plane.

proxy in our possibly non-convex setting. The qualitative findings—existence of asymmetric NEs, monotone coalition attack, externality lemma—do not depend on the precise shape of V; they require only that V is continuous, monotone-decreasing in σ, and bounded. We expect the same qualitative picture under any utility function with these three properties.

Mitigation. The natural mitigation is to make the contract internalise the externality. The $(\varepsilon_i, \delta, \Delta)$-iDP formulation of [8] adds a Δ-divergence bound on the realised vulnerability, restricting the action space of the PEG to profiles that satisfy a per-record bound on excess advantage. We expect this restriction to admit a unique equilibrium and to drive the realised advantage gap between players to within Δ by definition. A clean characterisation of the constrained PEG is left to future work.

Open questions. The following questions naturally arise from this work but lie outside its scope:

1) Does a pure-strategy NE always survive on the continuous action space under the coupled cost, and does a Δ-bound on the slack suffice for existence?
2) Is the PEG a potential game in the sense of [26]? PoP's analysis was a potential game because the privacy cost was separable; the externality term breaks this directly. Conditions on the cross-partial of A_i that recover the potential structure would unify the two settings.
3) What is the sign of the externality across composition regimes (minority vs majority weak-privacy)? Our experiments suggest the smaller group is favoured, but the analytic dependence on $|G_g|$ deserves a closed-form characterisation.
4) What is the appropriate player granularity (single sample versus silo) in the FL setting, and how does the granularity change the equilibria?
5) Worst-case (analytical) vs realised (empirical) cost—which should the contract be written against, and how large is the gap in practice across deep-learning workloads?
6) Is the $(\varepsilon_i, \delta, \Delta)$-iDP contract by itself sufficient as a collusion-resistant mechanism, or are sensitivity-based iDP or public-data padding needed when the slack is exceeded?

Wider lesson. The wider lesson is that an *individual* privacy contract is incomplete: it does not pin down the collective dynamics. Where individuals interact strategically (in a single machine-learning training, in a federated round, in any future composition of the iDP mechanism), the contract must also constrain their joint behaviour or the equilibrium will quietly violate at least one participant's expectations while preserving every formal guarantee.

Conclusion. The promise of iDP—that every data contributor controls her own privacy—is broken by the privacy externality of sampling-based mechanisms. We formalised the strategic situation faced by iDP data contributors as the Privacy Externality Game, proved monotonicity of the externality and characterised the equilibria. The result is that the collusion attack on iDP is not adversarial but *rational*, and the remedy must operate at the level of the contract.

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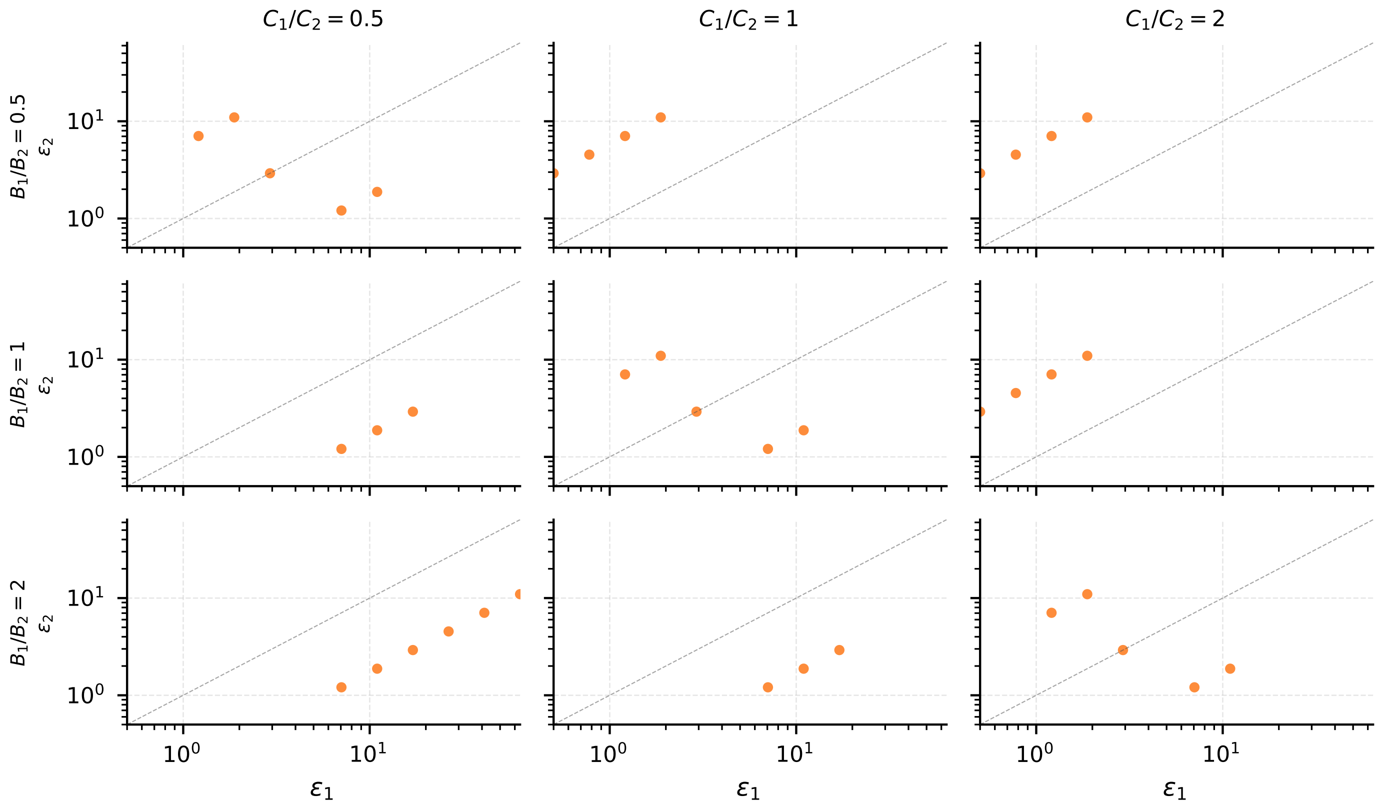

Fig. 8: **C9 (small multiples).** Pure NE locations in $(\varepsilon_1, \varepsilon_2)$ as the player asymmetries B_1/B_2 and C_1/C_2 are swept jointly. Each cell of the 3×3 grid shows the NEs at one $(B_1/B_2, C_1/C_2)$ combination; the dashed reference is the diagonal $\varepsilon_1 = \varepsilon_2$.

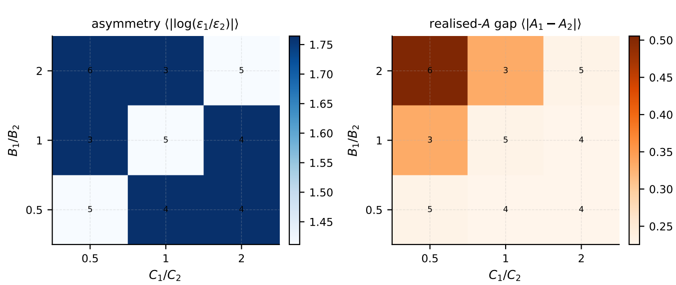

Fig. 9: **C9 (scalar summaries).** Two-scalar projection of the four-dimensional NE dependence: the mean log-asymmetry of the NE locations (left, blue) and the mean realised-vulnerability gap between the two players (right, orange). Annotations give the number of pure NEs in each cell.

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APPENDIX A
MECHANISM-SOLVER CONSTRUCTION

The mechanism solution $(\sigma, p_1, \dots, p_n)$ for a given action profile ε is obtained by a nested bisection. For each candidate σ, we find the largest per-group p_g that satisfies the I-fold (p_g, σ)-subsampled Gaussian at (ε_g, δ)-DP under (C1). Because the achieved ε is monotone-increasing in p_g, a unique $p_g \in (0, 1]$ exists (capped at 1 if the budget cannot be exhausted); we locate it by binary search using the Rényi-DP accountant [27] until the achieved ε matches ε_g to a relative tolerance of 10^{-3}. We then bisect σ on top of this inner search, using the realised batch size $B(\sigma) = \sum_g |G_g|\, p_g(\sigma, \boldsymbol{\varepsilon})$ as the outer-loop monitor, until B matches E_b to the same tolerance. Profiles in which any p_g falls below 10^{-5} are flagged infeasible per Theorem 4; below that threshold the group's contribution to (C2) is numerically negligible and the iDP contract is trivially satisfied. The qualitative results are insensitive to this threshold across $[10^{-6}, 10^{-3}]$.

Once (σ, p_g) is fixed, the realised MIA advantage of group g is computed exactly from the privacy-loss-distribution of the I-fold (p_g, σ)-subsampled Gaussian [22] by evaluating the trade-off function and (2). This matches the formulation of [5] and yields the tightest analytical MIA advantage for the resulting mechanism.

APPENDIX B
VALIDITY OF $V(\sigma) = 1/(1 + \sigma^2)$

[7] prove that for a 1-Lipschitz strongly-convex $\ell\colon \mathcal{X} \times \mathcal{D} \to \mathbb{R}$, the excess empirical risk of DP-ERM satisfies

$$\mathbb{E}[\hat{L}(\widehat{\theta}) - \hat{L}(\theta^*)] \;=\; O\!\left(\frac{d\,\log(1/\delta)}{n^2\,\varepsilon^2}\right).$$

Under the moments accountant of [20] this translates into a $\sigma^2 \cdot d/n^2$ dependence of the excess risk. Treating model utility as the negative of excess risk and renormalising to $[0, 1]$ gives $V(\sigma) \propto 1/(1 + \alpha\,\sigma^2)$ for a problem-dependent constant α; we set $\alpha = 1$ for simplicity. The bound assumes convexity, but the empirical literature on DP-SGD in deep learning [2], [20], [21] confirms the same qualitative direction. The equilibrium results that follow do not depend on the particular functional form: only on V being continuous, monotone-decreasing in σ, and bounded.

APPENDIX C
SENSITIVITY CHECKS

We re-ran the 2-player setting of Fig. 1 at $K = 44$ (twice the resolution along each axis) and obtained the same eight pure NEs at nearly identical locations; the asymmetric multiplicity is therefore not a grid artefact. Substituting $V(\sigma) = e^{-\sigma}$ for $V(\sigma) = 1/(1 + \sigma^2)$ shifts the NEs quantitatively—they sit at somewhat higher ε values because the exponential decays faster as σ grows—but the asymmetric structure is preserved. Lowering the participation threshold from 10^{-5} to 10^{-6} enlarges the feasible region and uncovers a few additional NEs on the new interior; raising it to 10^{-3} removes some boundary NEs while leaving the interior ones unchanged.